Chapter 1 Answers

January 19, 2024

1

(a): First, we show that if S is a boolean truth set, then (1) and (2) follow.
(1): We are to show that $\alpha \in S$ iff $\alpha_1 \in S$ and $\alpha_2 \in S$. The case where α is $X \wedge Y$ is just condition (1). By (4), $\sim(X \supset Y)$ iff $X \supset Y \notin S$. So, $\sim(X \supset Y) \in S$ iff $X \supset Y \notin S$ iff $X \in S$ and $Y \notin S$ (by (3)) iff $\sim Y \in S$, which proves the case of $X \supset Y$. In the third case, the formula is of the form $\sim(X \vee Y)$. $\sim(X \vee Y) \in S$ iff $X \vee Y \notin S$ (by (4)) iff $X \notin S$ and $Y \notin S$ (by (2)) iff $\sim X \in S$ and $\sim Y \in S$ (by (4)), which completes this case. In the last case, the formula is $\sim\sim X$. $\sim\sim X \in S$ iff $\sim X \notin S$ (by (4)) iff $X \in S$ (by (4)), which completes the proof in this case.
(2): We are to show that either $X \in S$ or $(\sim X) \in S$, but not both. By (4), $(\sim X) \in S$ iff $X \notin S$. In any case, this shows that either $X \in S$ and $(\sim X) \notin S$, or that $(\sim X) \in S$ and $X \notin S$ (by definition of iff), which is what we were trying to prove. (b): Next, suppose that the two conditions of Problem 1 hold. We are to show that S is a Boolean truth set. (1): This is immediate from condition (1) of Problem 1.
(2): By (2) from Problem 1 $X \vee Y \in S$ iff $\sim(X \vee Y) \notin S$ iff $\sim X \notin S$ or $\sim Y \notin S$ (by (1) of Problem 1) iff $X \in S$ or $Y \in S$ (by (2) of Problem 1)
(3): By (2) from Problem 1 $X \supset Y \in S$ iff $\sim(X \supset Y) \notin S$ iff $X \notin S$ or $\sim Y \notin S$ (by (1) of Problem 1) iff $\sim X \in S$ of $Y \in S$ (by (2) of Problem 1)
(4): This is obviously equivalent to (1) of Problem 1.

Exercise 1

(a): First, we show that if S is a boolean truth set, then (1) and (2) from this exercise follow.
(1): The cases where β is of type $X \vee Y$ or $X \supset Y$ are immediate (by (2) and (3) from the Boolean consequence relation definition). When β is of the form

$\sim(X \wedge Y)$, $\sim(X \wedge Y) \in S$ iff $X \wedge Y \notin S$ (by (4)) iff $X \notin S$ or $Y \notin S$ (by (1)) iff $\sim X \in S$ or $\sim Y \in S$ (by (4)).
(2): This is equivalent to (4) in the definition of a Boolean consequence relation.
(b): Next, we show that if S satisfies (1) and (2) from this exercise, then S is a Boolean truth set. (1): $X \wedge Y \in S$ iff $\sim(X \wedge Y) \notin S$ (by (2)) iff $\sim X \notin S$ and $\sim Y \notin S$ (by (1)) iff $X \in S$ and $Y \in S$ (by (2)).
(2-3): Conditions (2) and (3) are immediate from condition (1) of this exercise.
(4): This is equivalent to condition (2) of this exercise.

2

$$
\begin{aligned}
1: &\ S \vdash X\ (Hypothesis)\\
2: &\ S \vdash Y\ (Hypothesis)\\
3: &\ X, Y \vdash Z\ (Hypothesis)\\
4: &\ S, X, Y \vdash X, Y\ (C_1)\\
5: &\ S, X \vdash Y\ (2, C_2)\\
6: &\ S, X \vdash X, Y\ (5, 4, C_3)\\
7: &\ S \vdash X, Y\ (1, 6, C_3)\\
8: &\ S, X, Y \vdash Z\ (3, C_2)\\
9: &\ S \vdash Z\ (7, 8, C_3)
\end{aligned}
$$

3

We assume that S is consistent, and $S \vdash X$. We must show that S, X is consistent. Because S is consistent, this means that $S \vdash Z$ is not the for at least one formula Z. Now suppose that $S, X \vdash Z$ is the case. By C_3, this would mean that $S \vdash Z$, which is a contradiction. This means that $S, X \vdash Z$ is not the case, meaning S, X does not yield at least one formula Z, thus proving that it is consistent.

4

(a): We assume that S is consistent, $S \vdash X$ and $X \vdash Y$. By C_2, $S, X \vdash Y$ follows. By C_3, $S \vdash Y$ also follows, meaning S, Y is consistent by Problem 3.
(b): We assume that S is consistent, $S \vdash X$ and $S \vdash Y$ and $X, Y \vdash Z$. By Problem 2, $S \vdash Z$ follows, and thus, by Problem 3, S, Z is consistent.

5

(a): We assume that M is maximally consistent, $X \in M$ and $X \vdash Y$ and that M is maximally consistent. Suppose that $Y \notin M$. By C_1, $M \vdash X$, meaning, by Problem 4(a), that M, Y is consistent. If we assume that $Y \notin M$, this is in direct contradiction with the fact that M is maximally consistent, thus meaning $Y \in M$.
(b): We assume that M is maximally consistent, $X \in M$ and $Y \in M$ and $X, Y \vdash Z$. This means that $M \vdash X$ and $M \vdash Y$ by C_1. By Problem 4(b), it follows that M, Z is consistent. So, if $Z \notin M$, this is in direct contradiction with the fact that M is maximally consistent, meaning $Z \in M$.

Exericse 2

We need to show that $S \vDash X$ is a Boolean consequence relation, that is, that it satisfies $C_1 - C_8$.
C_1: Suppose $X \in S$. Because $X \in S$, X is obviously true under all interpretations that satisfy S, meaning $S \vDash X$.
C_2: Suppose $S_1 \vDash X$ and $S_1 \subseteq S_2$. This means that X is true under all interpretations that satisfy S_1. Well, because $S_1 \subseteq S_2$, all interpretations that satisfy S_2 also satisfy S_1, meaning X is also true under all interpreations that satisfy S_2, thus proving $S_2 \vDash X$.
C_3: Suppose $S \vDash X$ and $S, X \vDash Y$. This means that Y is true under all interpretations that satisfy S and satisfy X. But X is true under all interpreations that satisfy S, meaning, under all interpreations that satisfy S, S, X is also satisfied. Thus, $S \vDash Y$.
C_4: Suppose $S \vDash X$. Recall the Corollary to the Compactness Theorem for Propositional Logic that states that if X is a logical consequence of S then X is a logical consequence of some finite subset of S (logical consequence there means the same thing as tautological consequence here), which single-handedly proves C_4.
C_5: (a-b): By definition of an α formula, it is true iff α_1, α_2 are true, meaning that $\alpha \vDash \alpha_1, \alpha \vDash \alpha_2$.
C_6: Again, by definition of an α formula, it is true iff α_1, α_2 are true, meaning $\alpha_1, \alpha_2 \vDash \alpha$.
C_7: By definition, $X, \sim X$ is unsatisfiable. Hence, every formula Y is vacuously true under all interpreations that satisfy $X, \sim X$ (because there aren't any).
C_8: Suppose that $S, \sim X \vDash X$. This means that, if $\sim X$ was true under any interpretations that satisfied S, X would also be true, yielding a contradiction. Hence, $S \bigcup \sim X$ is unsatisfiable, meaning X is true under all interpretations that satisfy S.

6

We suppose that $\vdash$ is tautologically complete. This means that, whenever X is a tautological consequence of $\emptyset$. Because we are considering the empty set, this means that all elements of the empty set of vacuously satisfied by every interpretation. Because X is assumed to be a tautology, it to is satisfied by every interpretation, meaning X is a tautological consequence of $\emptyset$, and thus, because $\vdash$ is tautologically complete, $\emptyset \vdash X$.

7

We must prove that if $\vdash$ is a Boolean consequence relation and M is maximally consistent with respect to $\vdash$, then M is a Boolean truth set. Recall that because $\vdash$ is a Boolean consequence realtion, it satisfies $C_1 - C_8$. Also, because M is maximally consistent with respect to $\vdash$, it is not a proper subset of any consistent set with respect to $\vdash$ such that this set has an element not in M. We need to show that M is a Boolean truth set, which, by Problem 1, is the same as showing that it satisfies conditions (1) and (2) of that problem. As for condition (1), we need to show that $\alpha \in S$ iff $a_1 \in S$ and $a_2 \in S$. On the one hand, assume that $\alpha \in M$. By C_1, $M \vdash \alpha$. By C_5, $\alpha \vdash \alpha_1$ and $\alpha \vdash \alpha_2$. Hence, by two applications of Problem 5(a), we can conclude that $\alpha_1 \in M$ and $\alpha_2 \in M$. On the other hand, assume that $\alpha_1 \in M$ and $\alpha_2 \in M$. Again, by C_1, $M \vdash \alpha_1, M \vdash \alpha_2$. By C_6, $\alpha_1, \alpha_2 \vdash \alpha$. Then, by Problem 5(b), we can conclude that $\alpha \in M$.
Now, we need to show that M satisfies condition (2) of Problem 1. This condition states that either $X \in S$ or $(\sim X) \in S$, but not both. Suppose both X and $\sim X$ were in M. By C_7, $X, \sim X \vdash Y$. Because M is maximally consistent, it is consistent (with repsect to $\vdash$), meaning $M \nvdash Z$ for some formula Z. However, by C_7, $X, \sim X \vdash Z$, meaning, if $X, \sim X \in M$, then $M \vdash Z$ by C_2, which is a contradiction.
Next, we need to show that one and only one of $X, \sim X$ are in M. First, assume that $\sim X \notin M$. We need to show that $X \in M$. Because M is maximally consistent, $M, \sim X$ must be inconsistent (because $\sim X \notin M$). Hence, $M, \sim X \vdash Y$ holds for every Y, and in particular, $M, \sim X \vdash X$ holds, meaning, by C_8, $M \vdash X$ holds. By Problem 5(a), substituting M for X, it follows that $X \in M$. The contrapositive of this statement is if $X \notin M$ then $\sim X \in M$, which completes the proof.
Alternatively: To show that $M \vdash X$ means $X \in M$, one can also use C_3. In particular, recall that $M \nvdash Z$ for the Z defined above. Then, if $X \notin M$, this means M, X must be inconsistent (because M is maximally consistent), and thus $M, X \vdash Z$. By C_3, it would follow that $M \vdash Z$, which is a contradiction, meaning $X \in M$.

We can also prove the case where $X \notin M$ directly. For suppose $X \notin M$. Because we have shown that M satisfies condition (1) of Problem 1, it follows that $\sim\sim X \notin M$. Hence, because M is maximally consistent, $M, \sim\sim X$ is inconsistent, meaning $M, \sim\sim X \vdash \sim X$, and thus $M \vdash \sim X$ by C_8. Then, by Problem 5(a), $\sim X \in M$.

8

We are to show that for a Boolean consequence relation $\vdash$, consistency is compact. That is, we must show that a S is consistent with respect to $\vdash$ iff all finite subsets of S are also consistent with respect to $\vdash$.
Case 1: S is consistent:
In this case, we must show that all finite subsets F of S. Suppose that some finite subset F_1 of S is inconsistent, that is, $F_1 \vdash Z$ holds for every formula Z. Because S is consistent, there is some Z_1 such that $S \nvdash Z_1$. Because $\vdash$ is a Boolean consequence relation,

$$S, F_1 \vdash Z_1$$

holds by C_2. Because $F_1 \in S$, $S \vdash F_1$ by C_1, meaning $S \vdash Z_1$ holds by C_3, which is a contradiction, as we assumed $S \nvdash Z_1$, thus completing the proof in this case.
Case 2: All finite subsets F of S are consistent:
In this case, we must show that S is also consistent. Suppose S was inconsistent, that is, $S \vdash Z$ holds for every Z. Because every finite subset F of S is consistent, there is some formula Z for every F such that $F \nvdash Z$. Because S is inconsistent, we know that $S \vdash X$ and $S \vdash \sim X$ for every formula X. By C_4, $F_1 \vdash X$ and $F_2 \vdash \sim X$ for some finite subsets F_1, F_2 of S. Now consider $F = F_1 \bigcup F_2$. $F \in S$, as, if $X \subseteq S$ and $Y \subseteq S$, then $X, Y \subseteq S$. By C_1, $F \vdash F_1$ and $F \vdash F_2$, and therefore $F \vdash X$ and $F \vdash \sim X$, meaning, by problem 2, substituting $X, \sim X$ in for Z, we have $F \vdash X, \sim X$, which proves $F \vdash Z$ for every Z (by C_7), thus yielding a contradiction (as we assumed all finite subsets of S were consistent).

9

We are to prove Theorem 1, that is, every Boolean consequence relation is tautologically complete. That is, we must show that, for every set S and formula X, whenever X is a tautological consequence of S, then $S \vdash X$. Recall that every Boolean truth set has been shown to be satisfiable, thus meaning M is satisfiable. Because M is satisfiable, so is every subset of M, that is, every consistent set. Thus, every consistent set is satisfiable, meaning if a set is

unsatisfiable, it is inconsistent. Now, because X is a tautological consequence of S, $S, \sim X$ is unsatisfiable, meaning this same set is also inconsistent, meaning $S, \sim X \vdash X$ holds, and therefore, by C_8, $S \vdash X$ holds, completing the proof.

10

We suppose that $S \vdash X$ and $S \vdash X \supset Y$. Because $\vdash$ is assumed to be faithful to modus ponens, $X, X \supset Y \vdash Y$ holds, meaning, by Problem 2, $S \vdash Y$ holds.

11

(a): Suppose $X \supset Y$ is provable and $S \vdash X$. We are to show that $S \vdash Y$ is provable. Because $\emptyset \in S$, and $\emptyset \vdash X \supset Y$, $S \vdash X \supset Y$ holds by C_2, and then, by Problem 10, $S \vdash Y$.
(b): Suppose $X \supset (Y \supset Z)$ is provable, and $S \vdash X$ and $S \vdash Y$. We are to show that $S \vdash Z$. By Problem 11(a), $S \vdash Y \supset Z$ holds, and then, again by Problem 11(a), $S \vdash Z$ holds.
(c): Suppose $X \supset Y$ is provable. We are to show that $X \vdash Y$ is provable. By C_2, $X \vdash X \supset Y$ holds. Because $\vdash$ is faithfull to modus ponens, $X, X \supset Y \vdash Y$ holds, and $X \vdash X$ holds by C_1. By Problem 2, $X \vdash Y$ holds.
(d): Suppose $X \supset (Y \supset Z)$ is provable. We are to show that $X, Y \vdash Z$. Again, by C_2, $X, Y \vdash X \supset (Y \supset Z)$. By C_1, $X, Y \vdash X$. Thus, $X, Y \vdash X$ and $X, Y \vdash X \supset (Y \supset Z)$, meaning, because $X, X \supset (Y \supset Z) \vdash Y \supset Z$, $X, Y \vdash Y \supset Z$ by Problem 2. Then, because $\vdash$ is faithfull to modus ponenes, $Y, Y \supset Z \vdash Z$. By C_1, $X, Y \vdash Y$, meaning, by Problem 2, $X, Y \vdash Z$.

12

We are to show that $\vdash$ is a consequence relation, where $\vdash$ means that X is deducible from S. Thus, we need to show that $\vdash$ satisfies $C_1 - C_4$.
C_1: If $X \in S$, then X is deducible from S, as a deduction was defined such that every term X_i could be a member of S, meaning we could have, for example, just the deduction X.
C_2: We suppose $S_1 \vdash X$, and that $S_1 \subseteq S_2$. We need to show that $S_2 \vdash X$, that is X is deducible from S_2. This is obvious as, because X is deducible from S_1, we can use this same exact deduction to show that X is deducible from S_2, as every term X_i that is a member of S_1 is also a member of S_2.
C_3: We need to show that if $S \vdash X$ and $S, X \vdash Y$, then $S \vdash Y$. Suppose we

have the sequence $X_1, ..., X_n$ deducing X from S, and the deduction $Y_1, ..., Y_k$ deducing Y from S, X. Thus, the deduction $X_1, ..., X_n, Y_1, ..., Y_k$ deduces Y from S (because X is a formula).
C_4: We need to show that if $S \vdash X$, then $F \vdash X$ for some finite subset of X. Because $X_1, ..., X_n$ is a finite sequence, we take F to be the set $Y_1 \bigcup ... \bigcup Y_k$, the set of the k formulae being subsets of S used in the deduction of X from S. F is obviously a finite subset of S, meaning, because $F \vdash X$, the proof is complete.

13

We assume A_1 and A_2, and are to prove that $X \supset X$ is provable. By A_2,

$$(X \supset (Y \supset X)) \supset ((X \supset Y) \supset (X \supset X))$$

is provable. Also, $X \supset (Y \supset X)$ is provable by A_1. Thus, by Problem 11(a),

$$(X \supset Y) \supset (X \supset X)$$

is provable (that is, $\emptyset \vdash (X \supset Y) \supset (X \supset X)$). Taking $Y \supset X$ for Y,

$$(X \supset (Y \supset X)) \supset (X \supset X)$$

is provable, meaning, by A_1 and Problem 11(a),

$$X \supset X$$

is provable.
Alternatively: Because the system A obeys the rule of modus ponens, we can replace the spots in the above proof where it says Problem 11(a) by modus ponens.

14

(a): We are to show that if $S \vdash Y$ then $S \vdash X \supset Y$. This is immediate from Problem 11(a), taking $X \supset Y$ to be $Y \supset (X \supset Y)$ (which is provable by A_1).
(b): We are to show that, if $S \vdash X \supset (Z \supset Y)$ and $S \vdash X \supset Z$, then $S \vdash X \supset Y$. This is immediate from Problem 11(b), taking $X \supset (Y \supset Z)$ to be

$$(X \supset (Z \supset Y)) \supset ((X \supset Z) \supset (X \supset Y))$$

(which is provable by A_2).

15

We are to prove the Deduction Theorem, that is, that the properties A_1 and A_2 holding for a system A are sufficient to prove that A satisfies the deduction property, that is, if $S, X \vdash Y$, then $S \vdash X \supset Y$. Suppose $S, X \vdash Y$. Because $S \vdash X$ here is the relation that X is deducible from S, this means that Y is deducible from $S \bigcup X$. Thus, there is a deduction $Y_1, ..., Y_n$ such that $Y = Y_n$. Now consider the sequence $X \supset Y_1, ..., X \supset Y_n$. Showing that $S \vdash X \supset Y_i$ holds for every i suffices to complete the proof, as it would mean that $S \vdash X \supset Y_n$ would hold, that is, $S \vdash X \supset Y$ would hold. There are then four cases for what every Y_i could be. To show all cases, we assume that if $S \vdash X \supset Y_j$ for all $j < i$, and we must show that $S \vdash X \supset Y_i$ (this is complete induction):
Case 1-2: In the first case, Y_i is an axiom of A. In this case, the proof sequence

$$Y_i, Y_i \supset (X \supset Y_i), X \supset Y_i$$

suffices as $X \supset Y_i$ follows from $Y_i \supset (X \supset Y_i)$ and Y_i by modus ponens, and $Y_i \supset (X \supset Y_i)$ is just A_1. In the second case where $Y_i \in S$, the same proof sequence suffices, as Y_i can be included in this sequence.
Alternatively: If Y_i is an axiom of A, $\vdash Y_i$, meaning, by C_2, $S \vdash Y_i$, and thus $S \vdash X \supset Y_i$ by Problem 14(a). If $Y_i \in S$, $S \vdash Y_i$ by C_1, meaning, again by Problem 14(a), $S \vdash X \supset Y_i$.
Case 3: In this case, $Y_i = X$. Thus we are trying to show that $S \vdash X \supset X$. By Problem 13, $X \supset X$ is provable, meaning, by C_2, $S \vdash X \supset X$.
Case 4: In this case, Y_i comes from two terms of the sequence $Y_1, ..., Y_i - 1$ by modus ponens, meaning these earlier terms are $Z \supset Y_i$ and Z for some formula Z. By the induction assumption, $S \vdash X \supset Z$ and $S \vdash X \supset (Z \supset Y_i)$. By Problem 14(b), it follows that $S \vdash X \supset Y_i$, thus completing the proof.

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We need to show that if a system A satisfies conditions $A_1 - A_6$ for a given consequence relation $\vdash$, then $\vdash$ is a Boolean consequence relation. Thus, we must derive $C_5 - C_8$.
C_5: By A_3, $\alpha \supset \alpha_1$ is provable and $\alpha \supset \alpha_2$ is provable. In both cases, $\alpha \vdash \alpha_1$ and $\alpha \vdash \alpha_2$ by Problem 11(c).
C_6: By A_4, $\alpha_1 \supset (\alpha_2 \supset \alpha)$ is provable. By 11(d), $\alpha_1, \alpha_2 \vdash \alpha$.
C_7: By A_5, $X \supset (\sim X \supset Y)$ is provable, meaning, by 11(d), $X, \sim X \vdash Y$.
C_8: In this case, we suppose that $S, \sim X \vdash X$. By A_6, $(\sim X \supset X) \supset X$ is provable, meaning, by C_2, $S \vdash (\sim X \supset X) \supset X$. By the Deduction Theorem, $S \vdash \sim X \supset X$, and thus, by Problem 10, $S \vdash X$.
Alternatively: You can use Problem 11(a) instead of C_2 to reduce the proof of C_8 by one step.